

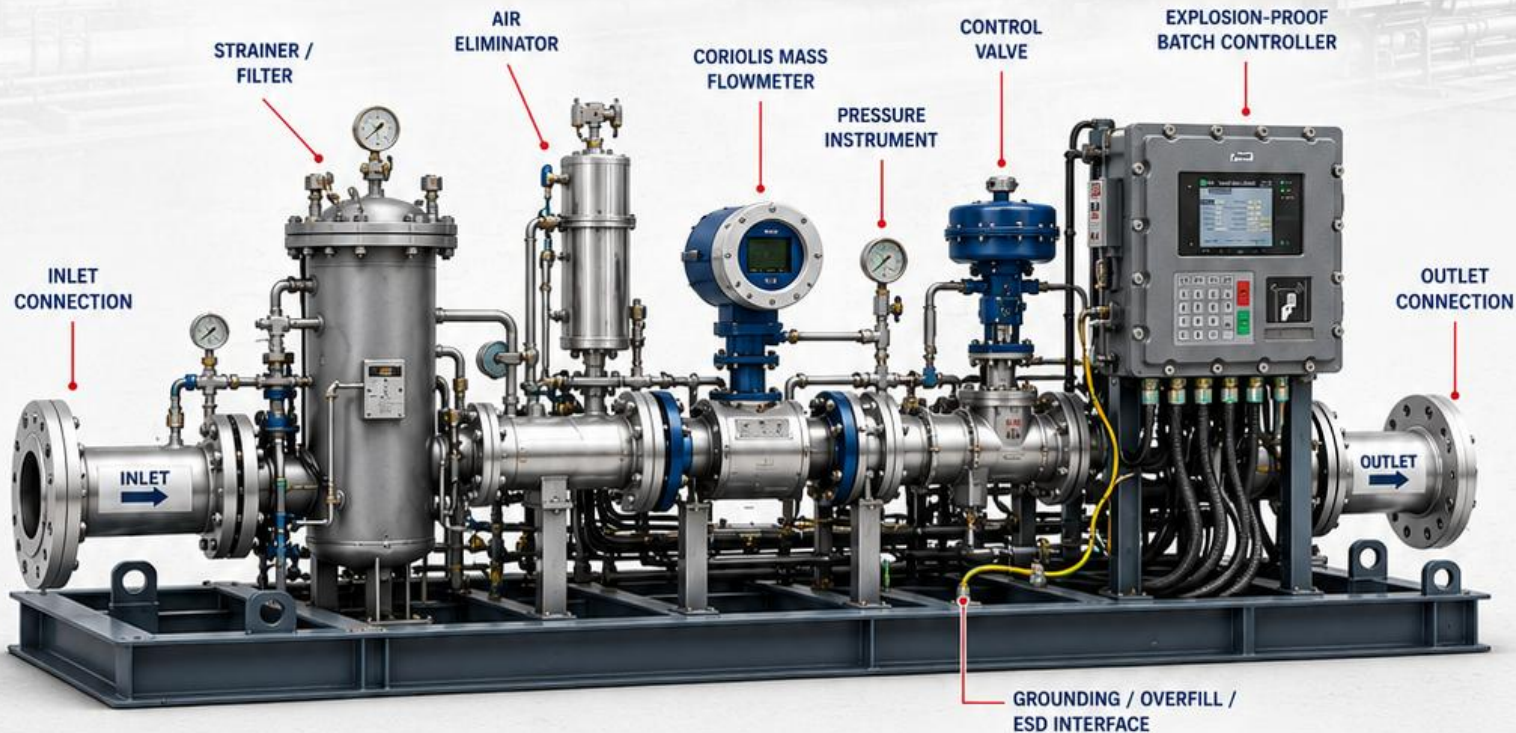
LNS-MS SERIES

Tanker Loading Metering Skids

TECHNICAL DATASHEET

Product Family: **LNS-MS**

Liquid Hydrocarbon | Fuel Oil & Base Oil | LPG | LNG



PRODUCT FAMILY MATRIX

MODEL	SERVICE DESCRIPTION	METERING TECHNOLOGY	TYPICAL APPLICATION
LNS-MS-LH	Refined & Light Hydrocarbon Loading	Coriolis Mass Flowmeter (Preferred)	Road Tanker / Depot / Terminal
LNS-MS-FO	Fuel Oil, Base Oil & Lube Oil Loading	Coriolis or PD Metering	Heated / High Viscosity Service
LNS-MS-LPG	LPG Loading Metering Skid	Coriolis Mass Flowmeter	Vapor Return / Safety / Ex-Proof
LNS-MS-LNG	LNG Loading Metering Skid	Cryogenic Coriolis	Cryogenic / Vapor Return / ESD



STANDARD SCOPE OF SUPPLY

- Inlet Isolation & Strainer / Filter
- Air Eliminator / Gas Separator
- Coriolis Mass Flowmeter
- Pressure & Temperature Instrumentation
- FCV / Batch Control Valve
- Explosion Proof Batch Controller
- ESD / Overfill / Grounding Interface
- Loading Arm / Hose Interface
- TAS / PLC / SCADA Integration



KEY DESIGN FEATURES

- Accurate Custody Transfer Ready Metering
- Coriolis Preferred Where Applicable
- Hazardous Area Ready Design
- Compact Skid-Mounted Construction
- Safe Batch Loading Logic
- Product-Specific Configuration for LH / FO / LPG / LNG
- Optional Heating / Tracing / Vapor Return / Cryogenic Design



APPLICATIONS

- Refineries
- Storage Terminals
- Fuel Depots
- Tank Farms
- Base Oil & Lube Terminals
- LPG / LNG Loading Stations
- Marine & Road Tanker Loading Systems



Custom-engineered metering skid solutions for tanker loading applications.



1. Product Family Overview

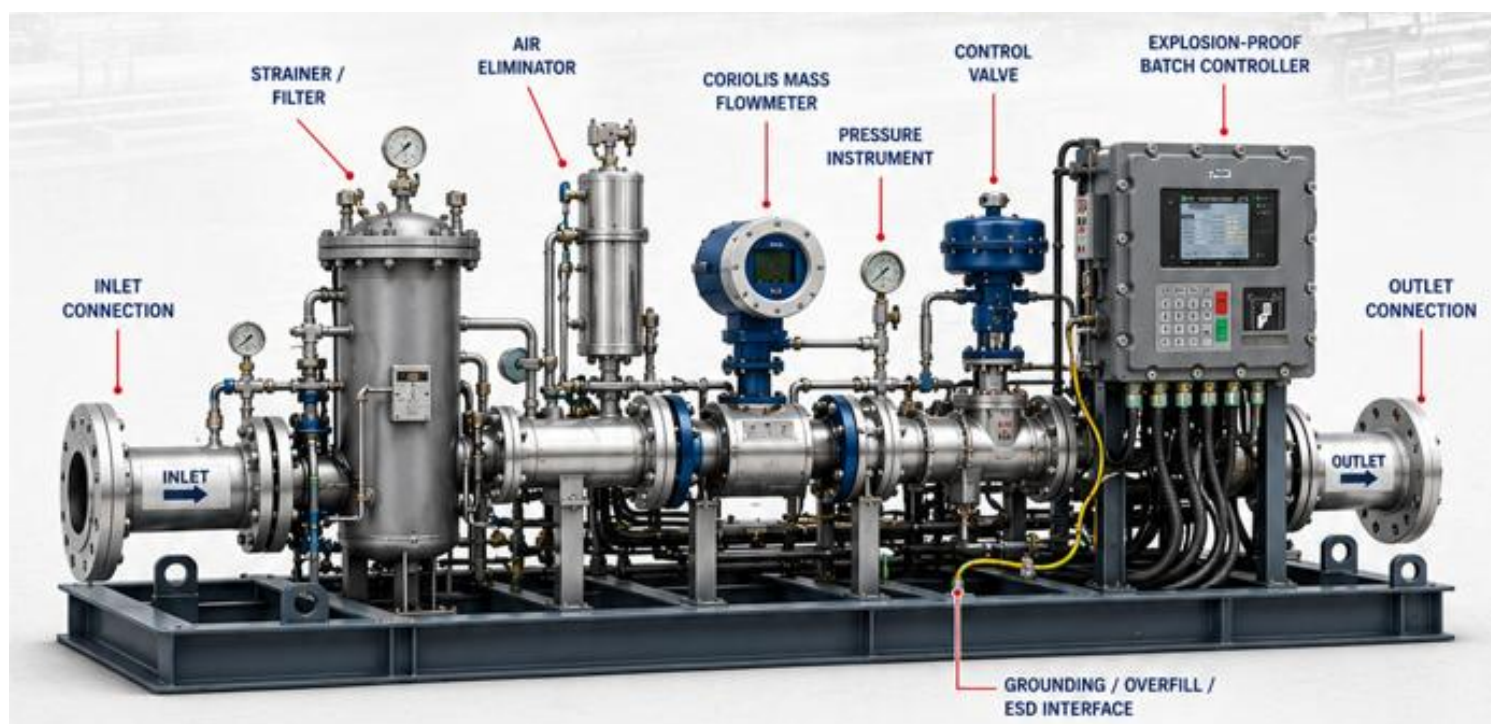
One scalable tanker-loading metering platform - four engineered service families.

The LNS-MS Series combines metering, flow conditioning, batch control, loading safety and terminal automation interfaces into a skid-mounted package. The family is intentionally divided by process behaviour rather than by cosmetic model differences: light hydrocarbons, high-viscosity oils, pressurised LPG and cryogenic LNG each require a different engineering approach.

Model	Service	Preferred Metering	Primary Engineering Driver	Typical Loading Environment
LNS-MS-LH	Refined & light liquid hydrocarbons	Coriolis mass flowmeter	Batch accuracy, clean flow, vapour control where applicable	Fuel terminals, depots, refineries, road/marine loading
LNS-MS-FO	Fuel oil, base oil & lube oil	Coriolis preferred; PD evaluated where justified	Viscosity, temperature, pressure drop and heating	Base-oil plants, lube terminals, heavy-oil depots
LNS-MS-LPG	Liquefied petroleum gas	Coriolis mass flowmeter	Maintain liquid phase, ESD, vapour/equalisation control	LPG terminals, filling plants, road/rail/marine loading
LNS-MS-LNG	Liquefied natural gas	Cryogenic Coriolis mass flowmeter	Cryogenic materials, cooldown, BOG and thermal movement	LNG loading/unloading and cryogenic transfer facilities

What stays common across the family

- Skid-mounted construction with product-specific piping, valves and instruments.
- Explosion-proof or hazardous-area batch control where required.
- Local and remote operating modes with PLC/TAS/SCADA interfaces.
- Grounding, overfill and emergency shutdown interfaces for tanker loading.
- Project-specific meter sizing from minimum, normal and maximum loading rates.



2. Common LNS-MS Architecture

The standard liquid-service train is deliberately simple, inspectable and maintainable.

Functional Stage	Typical Equipment	Engineering Purpose
Inlet isolation	Manual/MOV/ESDV as project requires	Positive isolation and safe maintenance
Flow conditioning	Strainer/filter; duplex filter option	Protect meter and downstream equipment
Gas removal	Air eliminator/gas separator where applicable	Prevent metering error from entrained gas in atmospheric liquid service
Pressure monitoring	PG + PT	Local indication, alarm and hydraulic diagnostics
Temperature monitoring	TE/TT where required	Product temperature and compensation/reporting input
Primary metering	Coriolis mass flowmeter	Direct mass measurement; density/diagnostics where supported
Flow/batch control	FCV, preset valve or batch shutoff valve	Rate control, ramping and batch termination
Local batch control	Explosion-proof batch controller	Preset, start/stop, totalisation, permissives, transaction record
Loading safety	Grounding, overfill, ESD, loading-arm/hose interface	Protect personnel, vehicle and terminal



Coriolis Mass Flowmeter



Explosion-Proof Batch Controller



Static Grounding Assembly

Important engineering distinction:

An air eliminator is appropriate for many atmospheric liquid-loading services, but LPG and LNG gas-management arrangements are different. Pressurised/cryogenic models require service-specific vapour and phase-control design rather than simply reusing the atmospheric liquid train.

3. LNS-MS-LH - Liquid Hydrocarbon Metering Skid

For gasoline/PMS, diesel/AGO, kerosene, Jet A1 and similar refined petroleum products.

LNS-MS-LH is the general-purpose high-accuracy loading platform for refined liquid hydrocarbons. The engineering objective is stable single-phase flow through the meter, controlled batch loading, safe vehicle interface and reliable transaction data.

Typical engineering configuration

Area	Recommended LNS-MS-LH Approach
Meter	Coriolis mass flowmeter preferred for direct mass and batch measurement
Filter/strainer	Sized to protect the meter with acceptable clean/dirty pressure drop
Air eliminator	Recommended where entrained air can enter from tanks/pumps/manifolds
Pressure instruments	PG + PT upstream/downstream as hydraulic study requires
Control valve	FCV or preset/batch valve with fail-safe action defined by HAZOP/C&E
PMS service	Vapour-return/VRU interface where required by loading philosophy
Jet A1	Dedicated clean-service review for filtration, materials, water control and sampling
Loading interface	Top or bottom loading depending product/fleet/terminal standard

Advantages

- Direct mass measurement and high repeatability for batch accountability.
- One compact skid integrates metering, control and loading permissives.
- Suitable for TAS-based order validation and transaction reporting.
- Easy to standardise across multiple loading bays with product-specific trim.

Applications

Refineries, product terminals, depots, airport-fuel terminals, marine terminals, tank farms and multi-product road-tanker gantries.



4. LNS-MS-FO - Fuel Oil, Base Oil & Lube Oil Metering Skid

The same metering objective, but a different hydraulic problem.

LNS-MS-FO is engineered around viscosity. Meter selection, pump differential pressure, control-valve sizing and line losses must be checked at the highest credible viscosity and lowest operating temperature - not only at normal warm conditions.

Engineering Item	LNS-MS-FO Approach	Why It Matters
Meter	Coriolis preferred where turndown and pressure drop are acceptable; PD may be evaluated	Viscous products can increase meter and line pressure loss
Temperature control	Steam/electric/thermal-oil tracing or jacketed components as required	Maintains viscosity within the loading design envelope
Insulation	Optional but frequently beneficial	Reduces heat loss and start-up delay
Strainer	Low-pressure-drop basket/duplex arrangement	Protects meter without excessive suction/discharge loss
Control valve	Oversizing avoided; Cv selected at hot and cold viscosity cases	Prevents poor control and excessive pressure drop
Recirculation/warm-up	Optional loop or bypass external to custody path if process requires	Allows safe preheating and line conditioning
Materials/seals	Selected for base oil/additives and maximum temperature	Ensures compatibility and service life

Operational advantages

- Stable loading across multiple oil grades when correctly temperature-conditioned.
- Reduced operator dependence through batch preset and controlled ramping.
- Viscosity-conscious hydraulic design lowers risk of slow loading and cavitation.
- Can be configured for base-oil blending, lube-oil dispatch or heated heavy-fuel transfer.

Metering selection: PD metering is not the default LINESIG position. Coriolis remains preferred where technically suitable; PD is retained as an engineering alternative for duties where line size, viscosity, pressure drop or project philosophy justify it.



5. LNS-MS-LPG - Pressurised LPG Metering Skid

Designed to keep LPG in the liquid phase through the metering and control train.

LPG loading requires more than a pressure-rated version of the liquid hydrocarbon skid. The complete hydraulic envelope must maintain adequate pressure above the product's flashing condition through filters, meter, valves and loading interface.

Engineering Item	LNS-MS-LPG Approach
Primary meter	Coriolis mass flowmeter for pressurised liquid LPG
Pressure control	Hydraulic design maintains single-phase liquid through the meter
Vapour handling	Vapour-return/equalisation connection where loading system requires it
ESD	Fail-safe ESD isolation integrated with batch and terminal shutdown logic
Grounding/overflow	Mandatory project safety interface for tanker loading
Breakaway/ERC	Available at the loading connection where specified
Pressure/temperature	Continuous monitoring for operation, diagnostics and transaction calculations
Hazardous area	Certified electrical equipment selected for the project zone/classification

Key advantages

- Direct mass measurement avoids dependence on volume correction alone.
- Compact package integrates metering, batch control and safety permissives.
- Suitable for road tanker, rail wagon and marine LPG loading architectures.
- Supports terminal automation and remote transaction reporting.

LINESIG LNS-MS SERIES – LPG CORIOLIS METERING SKID OPTIONS

For LPG (Liquefied Petroleum Gas) – Accurate, Safe & Reliable Custody Transfer

OPTION 1 – BASIC LPG METERING SKID



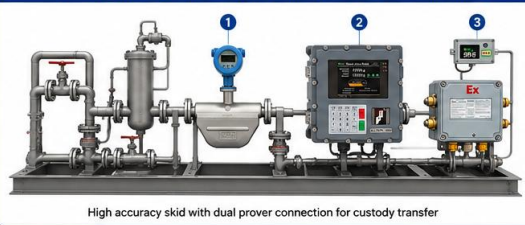
Compact design for truck loading / small flow applications

OPTION 2 – STANDARD LPG METERING SKID



Standard design for depot loading / unloading applications

OPTION 3 – ADVANCED LPG METERING SKID (DUAL PROVER CONNECTION)



High accuracy skid with dual prover connection for custody transfer

OPTION 4 – LPG VAPOR RECOVERY METERING SKID



With vapor return line for emission control during loading

COMPONENTS

1 CORIOLIS MASS FLOW
(As per attached image)



2 EXPLOSION PROOF BATCH CONTROLLER
(As per attached image)



3 STATIC GROUNDING ASSEMBLY
(As per attached image)



DESIGN FEATURES

- ATEX / IECEx Certified Equipment
- For Custody Transfer Applications
- High Accuracy Coriolis Technology
- Explosion Proof Batching Controller
- Static Grounding for Safe Operation
- Suitable for LPG (Liquid & Vapor)
- Skid Mounted, Pre-wired & Pre-tested

TYPICAL APPLICATIONS

- LPG Truck Loading
- LPG Depot Loading / Unloading
- LPG Cylinder Filling Plants
- LPG Bulk Storage Terminals
- LPG Marine / Rail Loading

Note: Piping, valves, strainer, pressure transmitter, temperature transmitter, and other accessories are included as per project specification.

ALL SKIDS CAN BE CUSTOMIZED AS PER CLIENT REQUIREMENTS

Phase management: LPG meter sizing is controlled by both flow range and allowable pressure loss. The selected Coriolis meter and FCV must be checked together with pump discharge pressure, vapour pressure and loading-arm losses.

6. LNS-MS-LNG - Cryogenic LNG Metering Skid

A dedicated cryogenic platform - not a material substitution of the ambient-temperature skid.

LNS-MS-LNG is engineered around cryogenic temperature, boil-off behaviour and thermal contraction. All wetted materials, valves, supports, seals and instrumentation must be qualified for the specified LNG temperature and loading duty.

Engineering Item	LNS-MS-LNG Approach
Primary meter	Cryogenic Coriolis mass flowmeter
Materials	Cryogenic stainless steel and qualified gaskets/seals
Piping	Insulated or vacuum-jacketed configuration according to project philosophy
Cooldown	Controlled cooldown/start-up sequence before stable custody/batch operation
BOG/vapour return	Dedicated boil-off/vapour-return interface
Valves	Cryogenic ESD/control valves with extended bonnet/stem arrangement as required
Thermal movement	Supports, anchors and flexibility designed for contraction/expansion
Emergency release	ERC/breakaway philosophy coordinated with loading arm/hose system
Detection	Gas detection and ESD interfaces coordinated with terminal safety system

Benefits

- Mass-based metering suits cryogenic transfer when the meter is selected for LNG service.
- Integrated package reduces site assembly and simplifies interface management.
- Skid architecture supports controlled cooldown, transfer and shutdown sequences.
- Modular automation allows interface to marine, truck or terminal LNG control systems.

LNG engineering focus

CRYOGENIC METERING

Coriolis sensor, transmitter, seals and cabling selected for the specified LNG temperature envelope.

BOG / VAPOUR RETURN

Boil-off and vapour-return paths are integrated with terminal LNG process and loading philosophy.

CONTROLLED COOLDOWN

Start-up sequence gradually establishes stable cryogenic conditions before normal metering and batch operation.

ESD / ERC INTEGRATION

Shutdown, gas detection and emergency-release functions are coordinated with the loading arm or hose.

PRE-COOL	READY	TRANSFER	BATCH COMPLETE	SAFE SHUTDOWN
Controlled cooldown	Stable liquid path	Coriolis metering + control	Close/ramp-down and totalise	Isolate, depressurise / warm-up per procedure

LNG design boundary: Final LNG design requires project-specific cryogenic process data, BOG philosophy, hazardous-area classification, loading-arm/ERC data and thermal-stress analysis. No generic ambient-service pressure/temperature values should be carried into this model.

7. Engineering Comparison - Selecting the Correct LNS-MS Model

The four models share a platform, but not the same process assumptions.

Design Topic	LNS-MS-LH	LNS-MS-FO	LNS-MS-LPG	LNS-MS-LNG
Fluid state	Ambient liquid	Viscous/heated liquid	Pressurised liquefied gas	Cryogenic liquefied gas
Meter	Coriolis preferred	Coriolis preferred / PD evaluated	Coriolis	Cryogenic Coriolis
Gas removal	Air eliminator often useful	Service-dependent	Phase control / vapour equalisation	BOG/vapour management
Temperature management	Usually ambient	Heating/tracing may govern	Maintain liquid phase	Cryogenic cooldown/insulation
Primary sizing driver	Flow/turndown/accuracy	Viscosity + pressure drop	Flash margin + pressure drop	Cryogenic duty + BOG + thermal effects
Typical loading	Top or bottom	Often top/bottom/side by product	Pressurised arm/hose	Cryogenic arm/hose/ERC
Special safety	Grounding/overfill/ESD	Grounding/overfill/ESD + hot surfaces	ESD + grounding + pressure/vapour safety	ESD + gas detection + ERC + cryogenic safety
Materials focus	Product compatibility	Temperature/viscosity compatibility	LPG compatibility/pressure class	Cryogenic stainless materials

Metering technology decision guide

Question	If YES	Engineering Direction
Is direct mass accountability required?	Yes	Coriolis strongly preferred
Is fluid viscosity very high at minimum temperature?	Yes	Check Coriolis pressure drop; evaluate PD if justified
Can liquid flash through the meter/control valve?	Yes	Increase pressure margin / resize equipment; LPG phase study required
Is service cryogenic?	Yes	Use cryogenic-rated Coriolis and LNG-specific skid design
Is custody transfer required?	Yes	Select certified meter/flow computer/proving/calibration architecture per project/regulatory basis

Quick service selector:

LNS-MS-LH Conventional refined liquids: prioritise clean single-phase flow, batch accuracy and loading interface.	LNS-MS-FO Viscous/heated liquids: prioritise cold viscosity, heating duty and total pressure loss.
LNS-MS-LPG Pressurised LPG: prioritise flash margin, vapour equalisation and fail-safe ESD.	LNS-MS-LNG Cryogenic LNG: prioritise cooldown, cryogenic materials, BOG and emergency release.

8. Metering, Batch Control & Terminal Automation

The skid is designed as a transaction system, not just a meter spool.

Function	Local Skid Function	TAS / PLC / SCADA Interface
Batch preset	Enter or receive target quantity	Order/batch download
Metering	Live mass/volume/density signals as configured	Transaction totals / live values
Pump demand	Start/stop request where permitted	Pump sequencing / permissive
FCV/batch valve	Ramp, regulate and terminate batch	Status / command / trip
Grounding	Healthy/unsafe permissive	Bay permissive and alarm
Overfill	Independent healthy/trip signal	Loading shutdown / event
ESD	Immediate stop input	Cause-and-effect and event logging
Transaction record	Batch quantity, time, operator/order reference	Historian / ticket / ERP interface

Typical batch sequence:

Validate order → Confirm product and route → Prove grounding/overfill/loading interface → Start pump demand → Open/ramp FCV → Meter and totalise → Slow-down/close at preset → Stop pump demand → Finalise transaction.

Automation options:

- Standalone explosion-proof batch controller.
- Batch controller with terminal TAS supervision.
- PLC-based control with local hazardous-area HMI.
- Redundant SCADA/TAS and historian integration for multi-bay terminals.
- RFID/card/vehicle identification, ticketing and ERP interface as project options.

Core transaction hardware:



Coriolis Meter



Explosion-Proof Batch Controller



Static Grounding Assembly

9. Safety, Interlocks & Loading Integrity

Safety permissives are designed around the loading operation, not around the meter alone.

Typical Permissive / Trip	Purpose	Typical Action
Grounding healthy	Prevents loading without verified tanker bond	Inhibit start; trip on loss
Overfill healthy	Independent tanker overfill protection	Inhibit start; close transfer valve on trip
Correct product/route	Prevents cross-contamination	Inhibit valve/pump sequence
Meter/controller healthy	Maintains controlled batch measurement	Inhibit or controlled stop
Pressure within limits	Detect blocked line/abnormal hydraulic condition	Alarm/trip as project requires
ESD pushbutton	Immediate emergency shutdown	Close FCV/ESDV and stop pump demand
Loading arm/hose ready	Prevents transfer before connection	Inhibit start
LPG/LNG vapour system ready	Protects pressurised/cryogenic loading sequence	Inhibit or shutdown per C&E

Hazardous-area philosophy

Electrical components are selected to the project's hazardous-area classification and certification regime. Batch controller, grounding unit, junction boxes, transmitters and actuators must be specified as a coordinated certified package; the exact certification marking is project- and market-specific.

Skid integrity

- Pressure-containing components rated to the project design pressure/temperature.
- Positive isolation for maintenance and safe depressurisation/draining provisions.
- Accessible instruments and valves for routine inspection and calibration.
- Bonding/earthing of skid, piping and electrical enclosures.

Three protection layers

 Batch / ESD Control		 Grounding / Overfill Interface
PROCESS PROTECTION Pressure monitoring, correct valve line-up, meter health and safe hydraulic operating envelope.	LOADING SAFETY Verified grounding, independent overfill protection, loading-arm/hose readiness and ESD.	CONTROL INTEGRITY Fail-safe shutdown logic, alarm/event recording and controlled operator/TAS permissions.

10. Options, Interfaces & Project Engineering

A standardised platform with project-specific engineering where it matters.

Option Group	Available Configuration
Metering	Coriolis; PD alternative on selected FO duties; proving/calibration connections
Filtration	Single basket, duplex, finer filtration, clean-service arrangement
Heating	Electric/steam/thermal-oil tracing, insulation, jacketed components
Additive / blending	Additive injection, ratio metering, recipe interface
Loading connection	Top arm, bottom arm, side arm, flexible hose, ERC/breakaway
Vapour	PMS vapour recovery, LPG equalisation, LNG BOG return
Control	Standalone batch controller, PLC/TAS, local HMI, remote I/O
Data	Printer, RFID/card, ticketing, historian, ERP interface
Materials	Carbon steel, 304/316L SS, cryogenic SS, project-specific metallurgy
Package layout	Single-line compact, extended maintenance layout, dual-meter/redundant arrangement

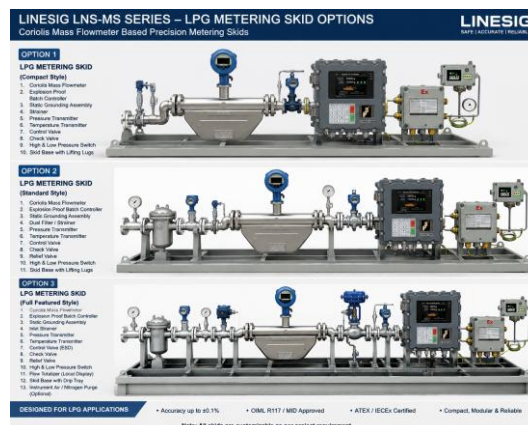
Project data required before final sizing

Input	Required Project Data
Product	Composition / grade / contamination limits
Flow	Minimum / normal / maximum loading rate
Batch	Typical batch size / tanker capacity / loading-time target
Hydraulics	Design/operating pressure, pump pressure, line losses, NPSH interface where relevant
Properties	Density, viscosity, vapour pressure, temperature range
Mechanical	Line size, flange standard/rating, materials/corrosion requirement
Electrical	Power supply, hazardous-area classification, certification basis
Control	TAS/PLC/SCADA protocol, I/O, batch/ticket requirements
Loading interface	Top/bottom/side arm, hose, vapour return, ERC/breakaway

Typical package layout choices



Extended maintenance / instrumentation layout



Compact service-specific arrangement

11. Model Selection & Ordering Guide

Choose the service family first; then size the package against actual process data.

Service Condition	Recommended Model	Key Selection Check
Refined fuels / light hydrocarbons	LNS-MS-LH	Flow range, clean service, vapour return, top/bottom loading
Fuel oil / base oil / lube oil	LNS-MS-FO	Cold viscosity, heating, meter/valve pressure drop
Propane/butane LPG	LNS-MS-LPG	Flash margin, pressure class, vapour equalisation, ESD
LNG / cryogenic liquid	LNS-MS-LNG	Cryogenic materials, cooldown, BOG, thermal movement, ERC

Proposed ordering structure

LNS-MS / SERVICE / SIZE / METER / CONTROL / LOADING / OPTIONS

Field	Typical Code	Description
SERVICE	LH / FO / LPG / LNG	Product-family model
SIZE	DN / NPS	Process line size
METER	CMF / PD	Coriolis mass flowmeter; PD only where approved
CONTROL	EX-BC / PLC / TAS	Local explosion-proof batch controller or integrated automation
LOADING	TOP / BOTTOM / SIDE / HOSE	Loading interface
OPTIONS	G / O / V / H / A / P / ERC	Grounding / Overfill / Vapour / Heating / Additive / Proving / Emergency release

LNS-MS-LH	LNS-MS-FO	LNS-MS-LPG	LNS-MS-LNG
LIGHT HYDROCARBON	FUEL / BASE OIL	PRESSURISED LPG	CRYOGENIC LNG

Why specify LINESIG LNS-MS

- One product family covering ambient liquid, viscous oil, pressurised LPG and cryogenic LNG loading duties.
- Coriolis-centred metering philosophy with service-specific engineering rather than one generic skid for every fluid.
- Integrated batch control, loading safety and terminal automation interfaces.
- Modular skid construction supports standardisation across multiple bays and future terminal expansion.
- Project-specific engineering keeps unsupported values out of the final datasheet until process conditions are confirmed.

Publication Note: This brochure describes the LNS-MS product-family engineering philosophy. Final performance, dimensions, meter accuracy, pressure class, certification and material selection are established by the project-specific technical datasheet and approved vendor/component data.

Meter → Batch Controller → TAS/PLC/SCADA → Historian / Ticket / ERP

LINESIG LNS-MS SERIES

Accurate Metering. Safe Loading. Reliable Control.